

```
programming class > 0526HW > C HW11-16.c > main()
1  #include <stdio.h>
2  #include <stdlib.h>
3  struct time {
4      int hour;
5      int minute;
6      double second;
7  };
8
9  int main() {
10     struct time start = {12, 32, 25.49};
11     struct time end = {15, 12, 17.53};
12     struct time elapse;
13
14     printf("start = %02d:%02d:%05.2f\n",
15         start.hour, start.minute, start.second);
16
17     printf("end = %02d:%02d:%05.2f\n",
18         end.hour, end.minute, end.second);
19
20     double start_total =
21         start.hour * 3600 +
22         start.minute * 60 +
23         start.second;
24
25     double end_total =
26         end.hour * 3600 +
27         end.minute * 60 +
28         end.second;
29
30     double diff = end_total - start_total;
31
32     elapse.hour = (int)(diff / 3600);
33     diff -= elapse.hour * 3600;
34
35     elapse.minute = (int)(diff / 60);
36     diff -= elapse.minute * 60;
37     elapse.second = diff;
38
39     printf("elapse= %02d:%02d:%05.2f\n",
40         elapse.hour, elapse.minute, elapse.second);
41     return 0;
42 }
```

```
double end_total =
    end.hour * 3600 +
    end.minute * 60 +
    end.second;

double diff = end_total - start_total;

elapse.hour = (int)(diff / 3600);
diff -= elapse.hour * 3600;

elapse.minute = (int)(diff / 60);
diff -= elapse.minute * 60;
elapse.second = diff;

printf("elapse= %02d:%02d:%05.2f\n",
    elapse.hour, elapse.minute, elapse.second);
return 0;
}
```

F:\Yuntech\programming class\0526HW\HW11-16.exe
start = 12:32:25.49
end = 15:12:17.53
elapse= 02:39:52.04

F:\Yuntech\programming class\0526HW>

```
programming class > 0526HW > C HW11-12.c > main()
1  #include <stdio.h>
2  #include <stdlib.h>
3  #define Max (2)
4
5  struct data {
6      char name[10];
7      int math;
8  }student[Max];
9
10 int main() {
11     int i;
12     struct data *p=student;
13     for(i=0; i<Max; i++) {
14         printf("Student name: ");
15         scanf("%s", (p+i)->name);
16         printf("Math score: ");
17         scanf("%d",&(p+i)->math);
18         while(getchar() != '\n');
19     }
20     for(i=0; i<Max; i++)
21         printf("%s's math score is %d\n",(p+i)->name, (p+i)->math);
22 }
```

問題 輸出 偵錯主控台 終端機 連接埠

F:\Yuntech\programming class\0526HW\HW11-12.exe
Student name: A
Math score: 100
Student name: B
Math score: 99
A's math score is 100
B's math score is 99

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  #define Max 5
4
5  struct data {
6      char name[10];
7      int math;
8  } student[Max];
9
10 struct data best(struct data student[]) {
11     struct data max = student[0];
12
13     for(int i = 1; i < Max; i++) {
14         if(student[i].math > max.math)
15             max = student[i];
16     }
17     return max;
18 }
19
20 void failed(struct data student[]) {
21     for(int i = 0; i < Max; i++) {
22         if(student[i].math < 60) {
23             printf("%s failed with %d\n",
24                 student[i].name,
25                 student[i].math);
26         }
27     }
28 }
29

```

```

double average(struct data student[]) {
    int sum = 0;

    for(int i = 0; i < Max; i++) {
        sum += student[i].math;
    }

    return (double)sum / Max;
}

void sort(struct data student[]) {
    struct data temp;

    for(int i = 0; i < Max - 1; i++) {
        for(int j = i + 1; j < Max; j++) {
            if(student[i].math < student[j].math) {
                temp = student[i];
                student[i] = student[j];
                student[j] = temp;
            }
        }
    }
}

```

```

int main() {
    for(int i = 0; i < Max; i++) {
        printf("Name: ");
        scanf("%9s", student[i].name);

        printf("Math: ");
        scanf("%d", &student[i].math);
    }

    struct data top = best(student);
    printf("\nBest student: %s %d\n", top.name, top.math);

    printf("\nFailed students:\n");
    failed(student);

    printf("\nAverage = %.2f\n", average(student));

    sort(student);

    printf("\nAfter sorting (high → low):\n");
    for(int i = 0; i < Max; i++) {
        printf("%s %d\n", student[i].name, student[i].math);
    }

    return 0;
}

```

```

Math: 30
Name: D
Math: 40
Name: E
Math: 50

```

Best student: E 50

Failed students:

```

A failed with 10
B failed with 20
C failed with 30
D failed with 40
E failed with 50

```

Average = 30.00

After sorting (high → low):

```

E 50
D 40
C 30
B 20
A 10

```